

Bug Tracking Report

Header Reference: Nishanth T Chigilar_PES1UG24CS302-BPS#56

Submission Metadata

- Student Name: Nishanth T Chigilar
- SRN: PES1UG24CS302
- Problem Statement: #56 — Peer Skill Exchange & Mentorship Network
- Course: Software Engineering Lab (UE24CS252AA) — Lab 2: Bug Tracking Deliverable
- Project Key: BT56

1. Objective & Defect Management Methodology

The objective of this module is to implement a structured Defect Tracking and Incident Management workflow using Atlassian Jira. During continuous integration and quality assurance testing of Problem Statement #56 (Peer Skill Exchange & Mentorship Network), functional anomalies, edge-case race conditions, and boundary validation failures were uncovered.

A dedicated Bug Tracking project (**BugTracker_BPS#56**) was established to capture defects with severity ratings, steps to reproduce, environmental contexts, expected versus actual behaviors, and resolution lifecycles. Defects progress across standard resolution states:

- **To Do:** New defects confirmed by QA awaiting investigation and root-cause analysis.
- **In Progress:** Defects under active remediation and patch development.
- **Done:** Patched issues that have been re-tested, validated, and verified as closed.

2. Bug Tracking Board Overview

The Jira Bug Tracking board tracks defect items distributed across workflow stages, prioritizing critical transaction flaws over UI boundary exceptions.

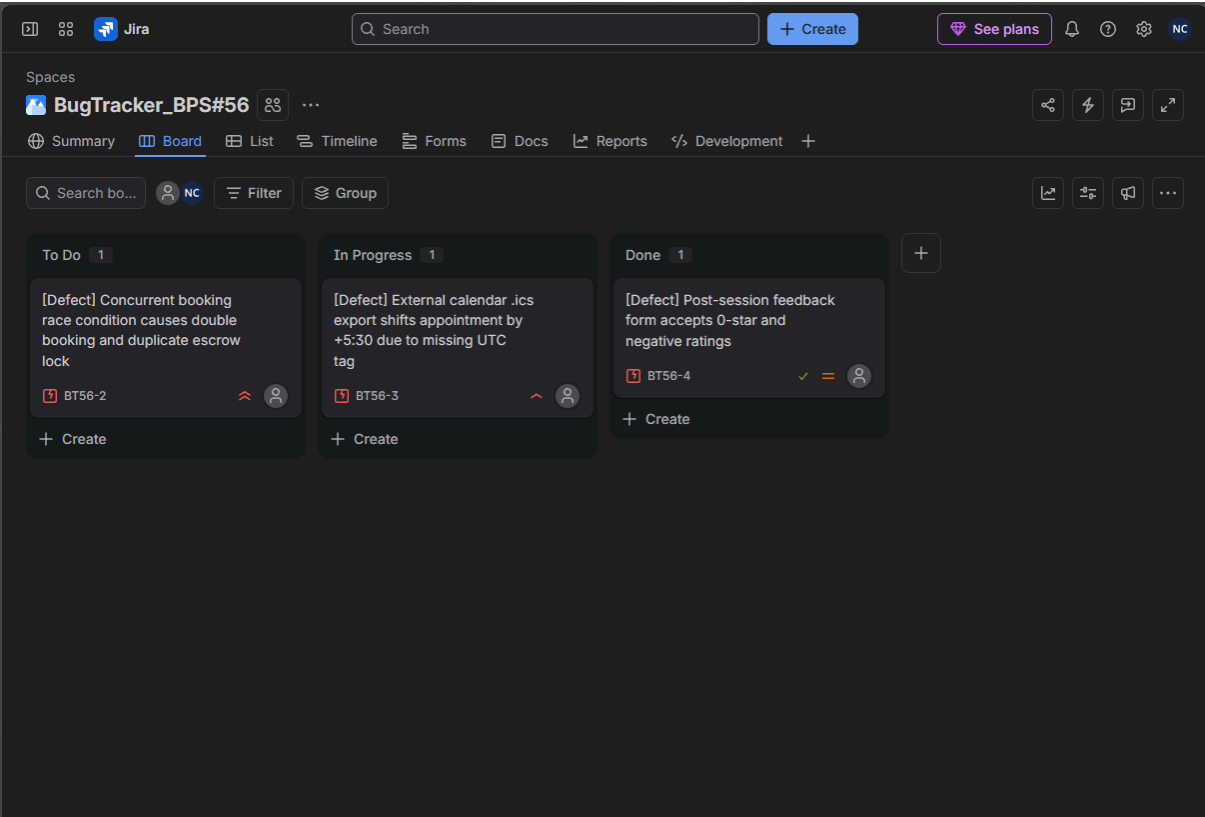


Figure 1: Jira Bug Tracking Board (BugTracker_BPS#56) displaying logged defects, priority indicators (Highest ↑↑, High ↑, Medium =), and workflow distribution across To Do (1), In Progress (1), and Done (1).

Logged Defects Summary

Issue Key	Defect Summary	Severity / Priority	Workflow State	Target Requirement
BT56-2	[Defect] Concurrent booking race condition causes double booking and duplicate escrow lock	Highest (↑↑)	To Do	FR-003 & NFR-002
BT56-3	[Defect] External calendar .ics	High (↑)	In Progress	NFR-001

Issue Key	Defect Summary	Severity / Priority	Workflow State	Target Requirement
	export shifts appointment by +5:30 due to missing UTC tag			
BT56-4	[Defect] Post-session feedback form accepts 0-star and negative ratings	Medium(=)	Done	FR-004

3. Detailed Defect Incident Reports

Defect Report 1: BT56-2

- **Summary:** [Defect] Concurrent booking race condition causes double booking and duplicate escrow lock
- **Issue Type:** Bug
- **Priority:** Highest (Blocker)
- **Status:** To Do / Open
- **Impacted Module:** Session Booking & Time-Banking Ledger (FR-003, NFR-002)

Description:

When two learners attempt to book the same mentor timeslot at the exact same sub-second interval, both transactions bypass available slot verification concurrently. The backend fails to enforce pessimistic row-level locking on the mentor schedule table, resulting in two confirmed sessions for a single time slot and duplicate escrow debit locks.

Steps to Reproduce:

1. Open two concurrent browser sessions logged into separate learner accounts.
2. Navigate to the same mentor profile and select an identical 1-hour open slot.

3. Dispatch simultaneous "Confirm Booking" requests within the same second.

Expected Result:

The first request acquires a database lock, reserving the slot and holding 1 credit in escrow. The second request must fail with an HTTP 409 Conflict status code and a user prompt: "This timeslot has just been reserved by another member".

Actual Result:

Both transactions commit successfully. Two booking confirmations are generated for the mentor, and 1 credit is escrowed from both learners for the same calendar block.

Recommended Fix:

Implement database row-level locking (SELECT ... FOR UPDATE) or transactional distributed locking within the reservation service to ensure atomic slot reservation.

Defect Report 2: BT56-3

- **Summary:** [Defect] External calendar .ics export shifts appointment by +5:30 due to missing UTC tag
- **Issue Type:** Bug
- **Priority:** High
- **Status:** In Progress
- **Impacted Module:** External Calendar Synchronization (NFR-001)

Description:

Upon confirming a mentorship session, exporting the session invite as an .ics file for Google Calendar or Microsoft Outlook results in an incorrect appointment start time. The timestamp is exported without standard timezone parameters, causing clients to misinterpret IST (Indian Standard Time, UTC+5:30) as UTC and shifting the meeting 5 hours and 30 minutes into the future.

Steps to Reproduce:

1. Book a mentorship session scheduled for 10:00 AM IST.
2. Click the "Export to Calendar (.ics)" button.
3. Open the downloaded .ics file in Google Calendar or Microsoft Outlook.

Expected Result:

The calendar event imports at exactly 10:00 AM IST with an explicit timezone identifier (TZID=Asia/Kolkata) conforming to RFC 5545 specifications.

Actual Result:

The meeting schedule displays at 3:30 PM IST (a +5:30 drift) because the generator omitted the Z UTC designator or explicit TZID parameter.

Recommended Fix:

Update the iCalendar serialization service to format timestamps in standardized UTC format (YYYYMMDDTHHMMSSZ) or explicitly append **TZID=Asia/Kolkata** to **DTSTART** and **DTEND** properties.

Defect Report 3: BT56-4

- **Summary:** [Defect] Post-session feedback form accepts 0-star and negative ratings
- **Issue Type:** Bug
- **Priority:** Medium
- **Status:** Done / Resolved
- **Impacted Module:** Session Feedback & Ratings Engine (FR-004)

Description:

The post-session review modal does not enforce boundary validation on star ratings prior to submission. Submitting the form with no star selected sends a payload with **rating: 0**, and manual payload intercept allows negative integer values, skewing mentor cumulative rating calculations.

Steps to Reproduce:

1. Complete a verified mentorship session.
2. In the "Session Feedback" dialog, leave star ratings unselected and enter a text comment.
3. Click "Submit Review".

Expected Result:

Client-side and server-side validation block submission, displaying an inline error: *"Please select a rating between 1 and 5 stars"*.

Actual Result:

Form submits successfully with **rating: 0**, causing the mentor's aggregate score to drop unfairly.

Verification & Resolution:

Patched and Verified. Schema validation was applied at the API gateway layer enforcing **`Joi.number().integer().min(1).max(5).required()`**, and frontend submit buttons are now disabled until a star selection is made. Ticket moved to Done.

4. Traceability Matrix & QA Summary

Logging defects directly against functional and non-functional requirements provides full traceability across the development lifecycle:

- **Data Integrity (NFR-002):** Identifying Defect BT56-2 protected the core time-banking ledger from silent balance corruption and multi-allocation vulnerabilities.
- **Interoperability (NFR-001):** Resolving Defect BT56-3 ensured adherence to open calendar standards, preventing missed mentor-learner sessions.
- **User Experience & Reputation (FR-004):** Closing Defect BT56-4 ensured the community rating system remains protected against erroneous inputs.